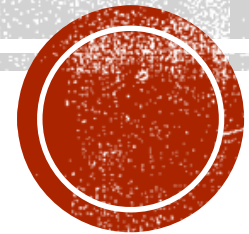


INTRODUCTION TO RECONFIGURABLE COMPUTING

CS G553: Reconfigurable computing

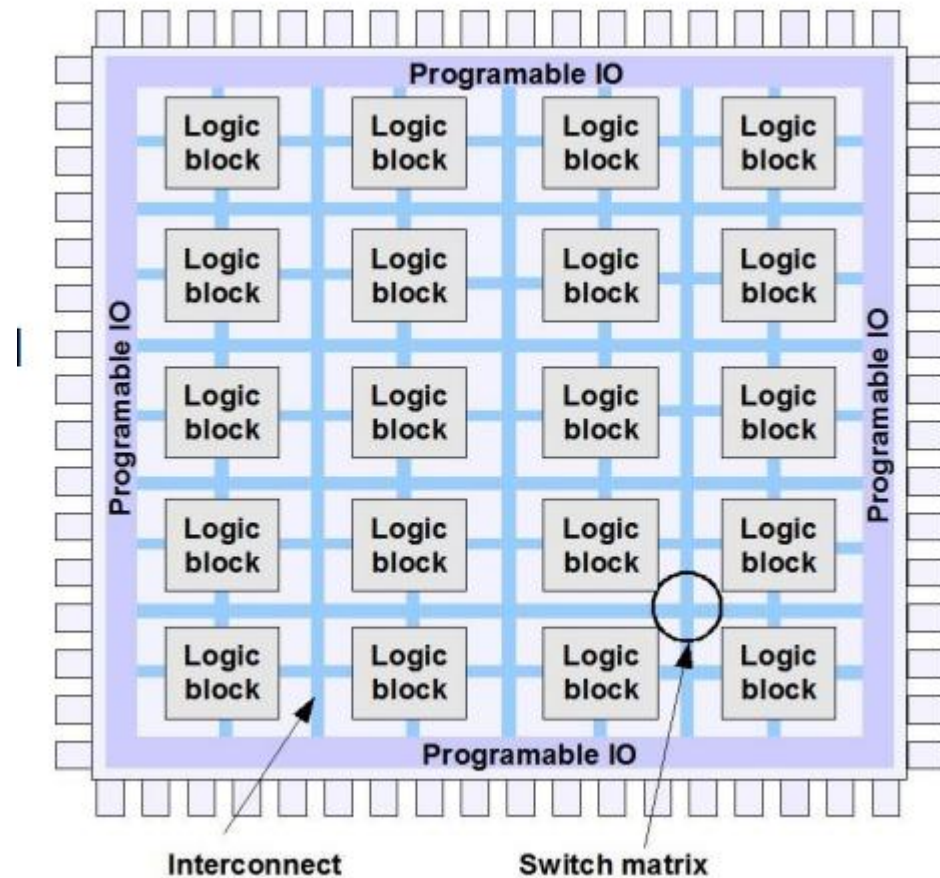
Dr. Meetha V Shenoy

Lect-5



FIELD PROGRAMMABLE GATE ARRAYS (FPGAs)

Introduced in 1985 by Xilinx

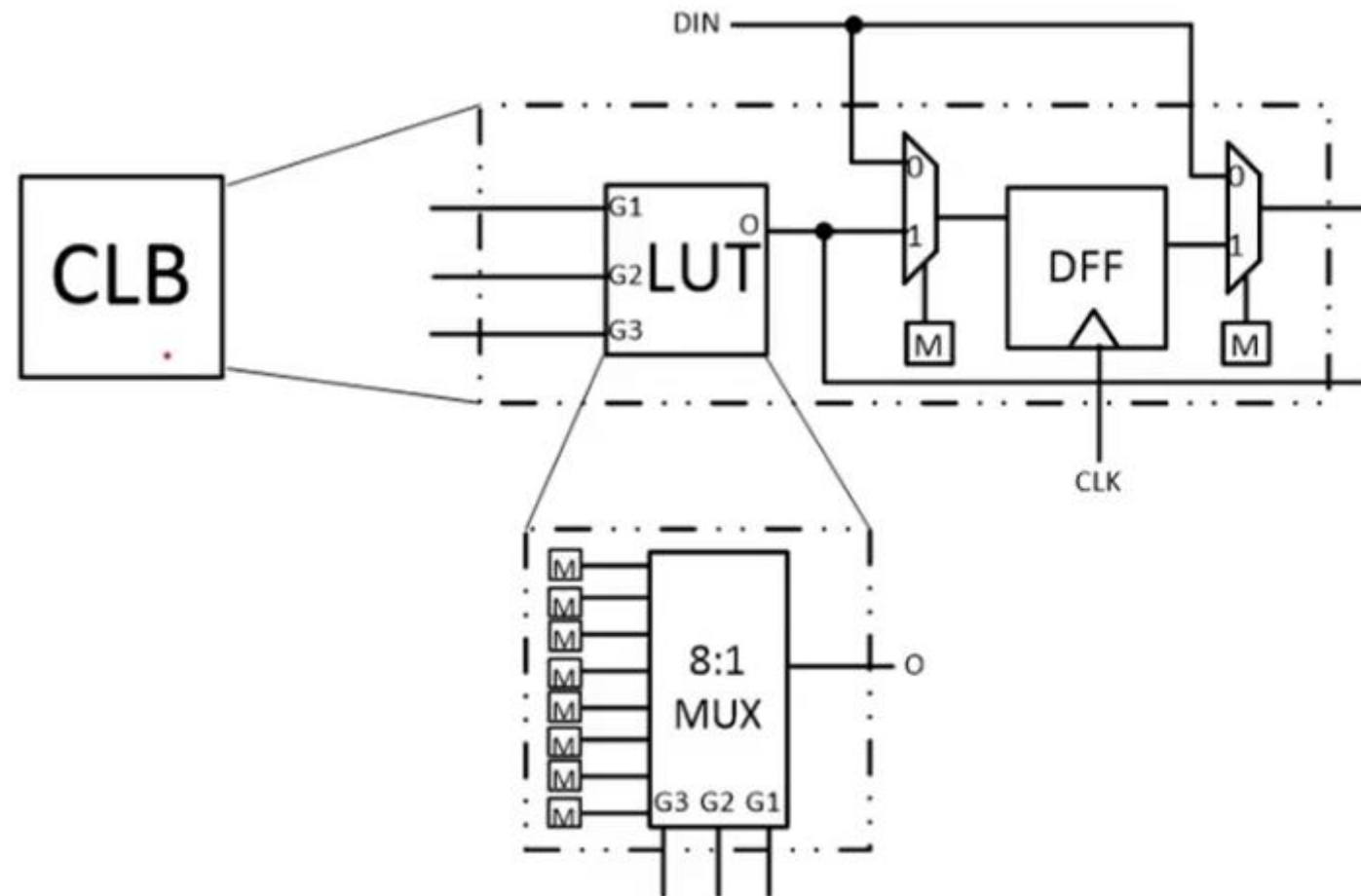


ELEMENTS OF FPGA

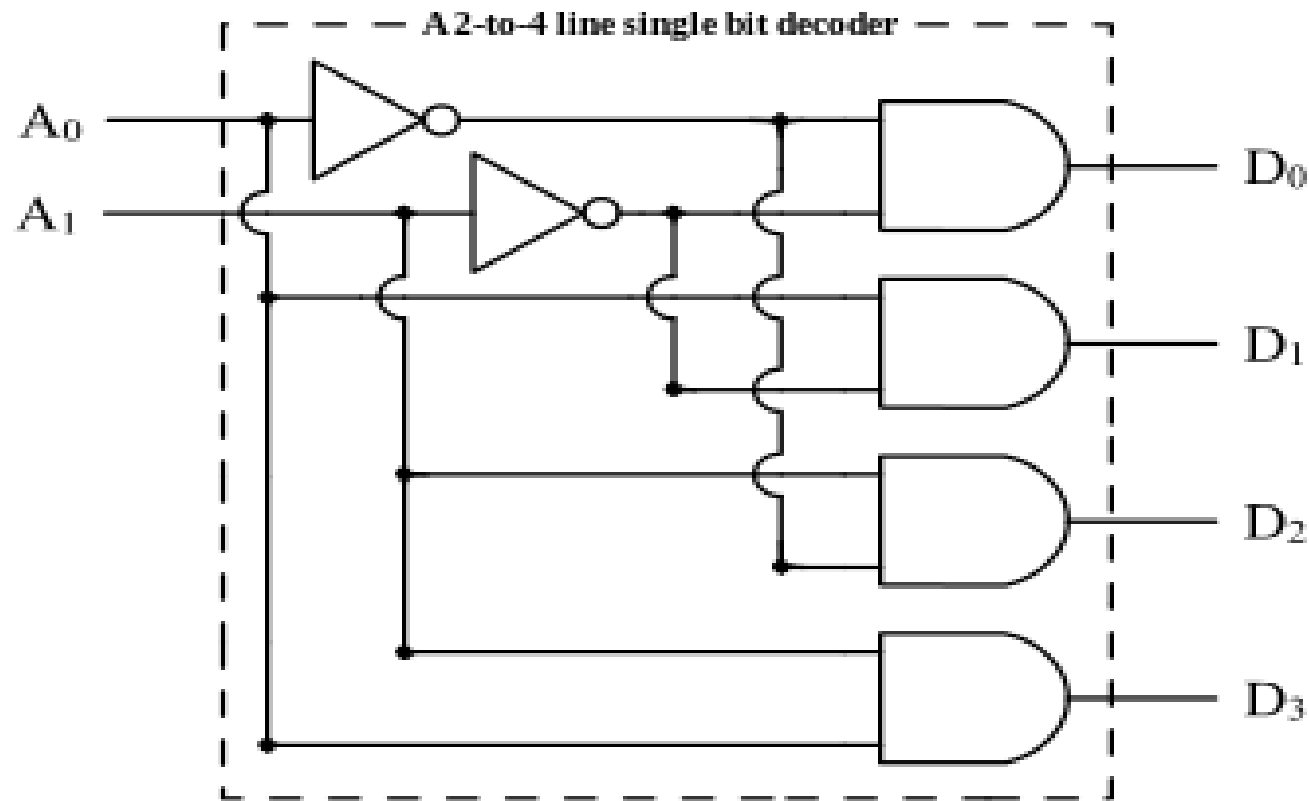
- Combinational/Sequential Logic
 - LUT: Look up table- Memory + Multiplexor
 - CLB/ Logic Element/Function Block/Logic Block /ALM: Configurable Logic Block/Adaptive Logic Module
- Interconnect
- IO Blocks



CLB ARCHITECTURE



IMPLEMENT 2X4 DECODER USING 3-INPUT 2-OUTPUT LUTS



Truth Table

A_1	A_0	D_3	D_2	D_1	D_0
0	0	0	0	0	1
0	1	0	0	1	0
1	0	0	1	0	0
1	1	1	0	0	0

Minterm Equations

$$D_0 = \overline{A_1} \cdot \overline{A_0}$$

$$D_1 = \overline{A_1} \cdot A_0$$

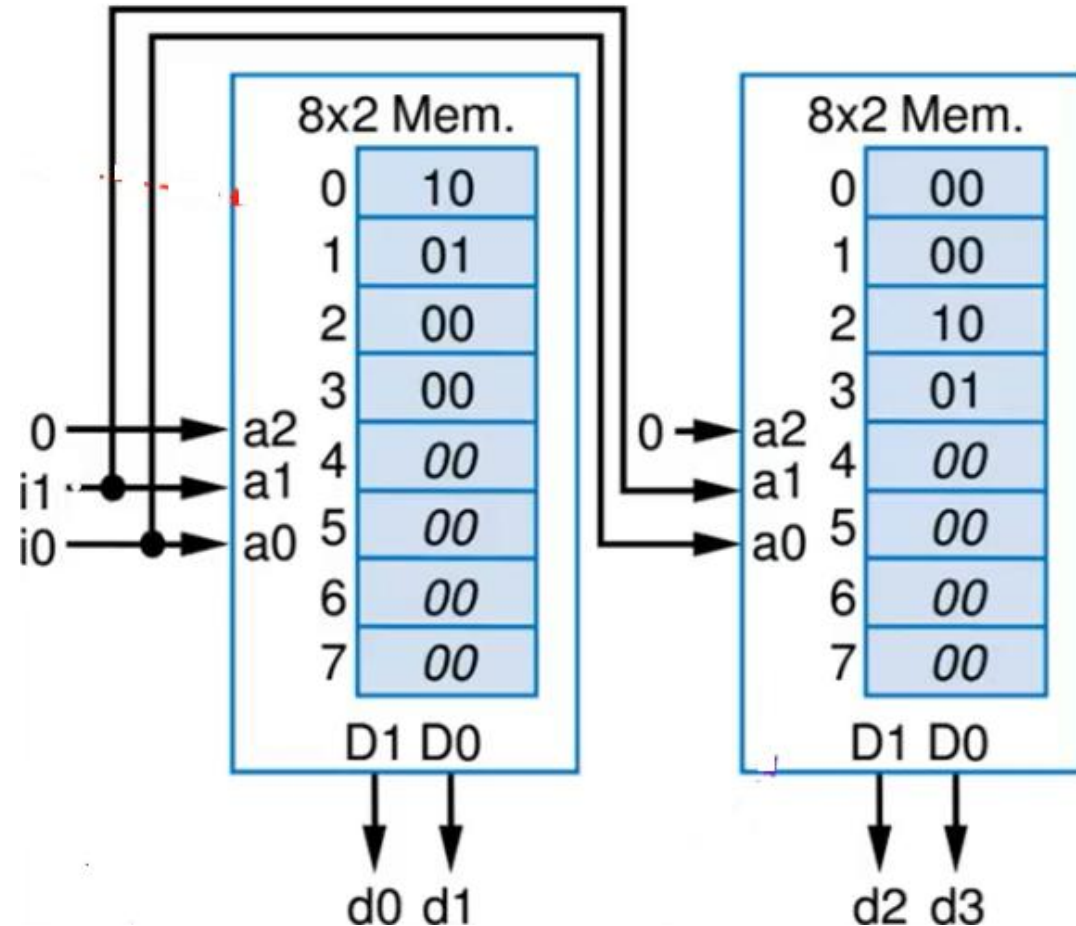
$$D_2 = A_1 \cdot \overline{A_0}$$

$$D_3 = A_1 \cdot A_0$$

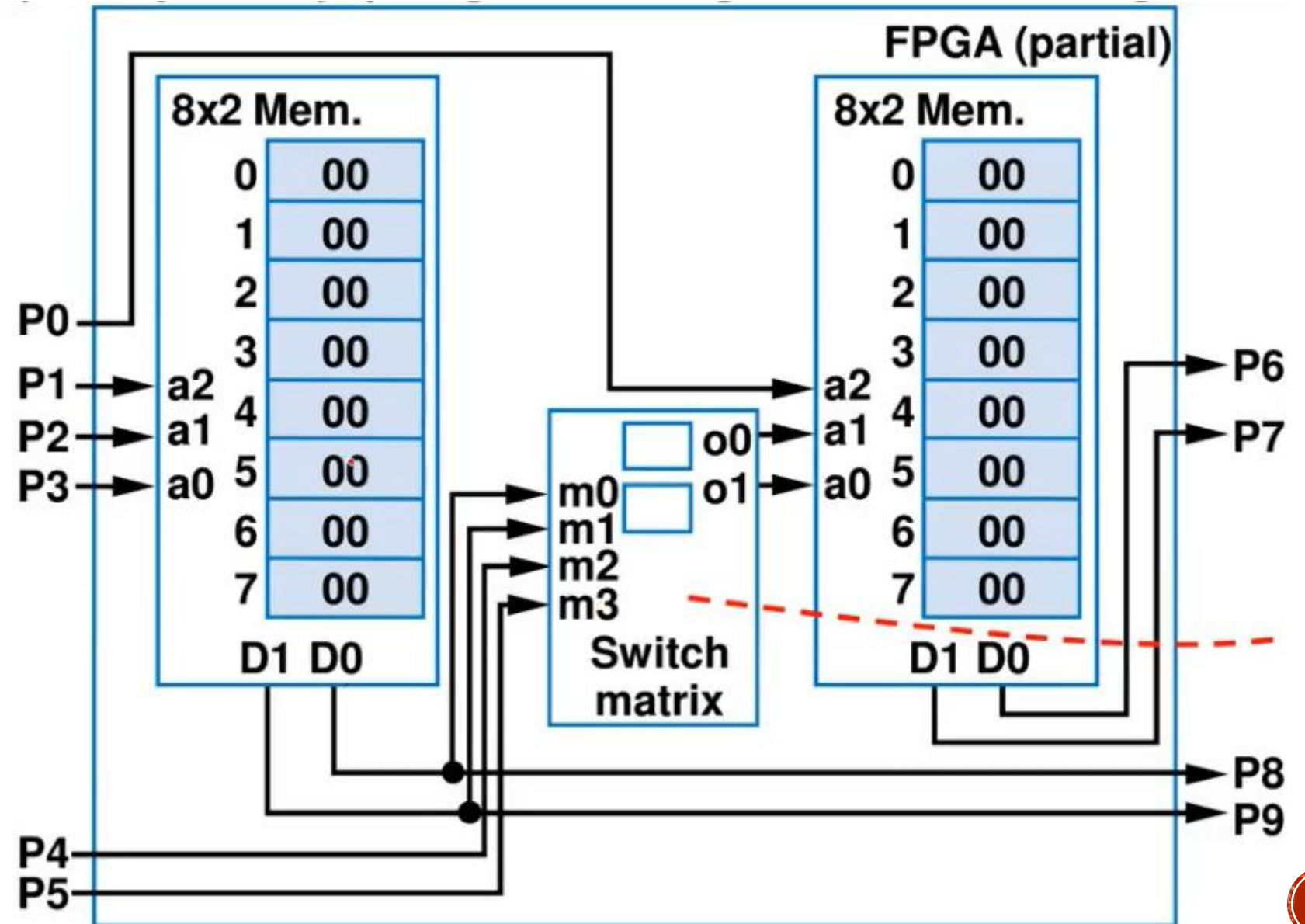


IMPLEMENT 2X4 DECODER USING 3-INPUT 2-OUTPUT LUTS

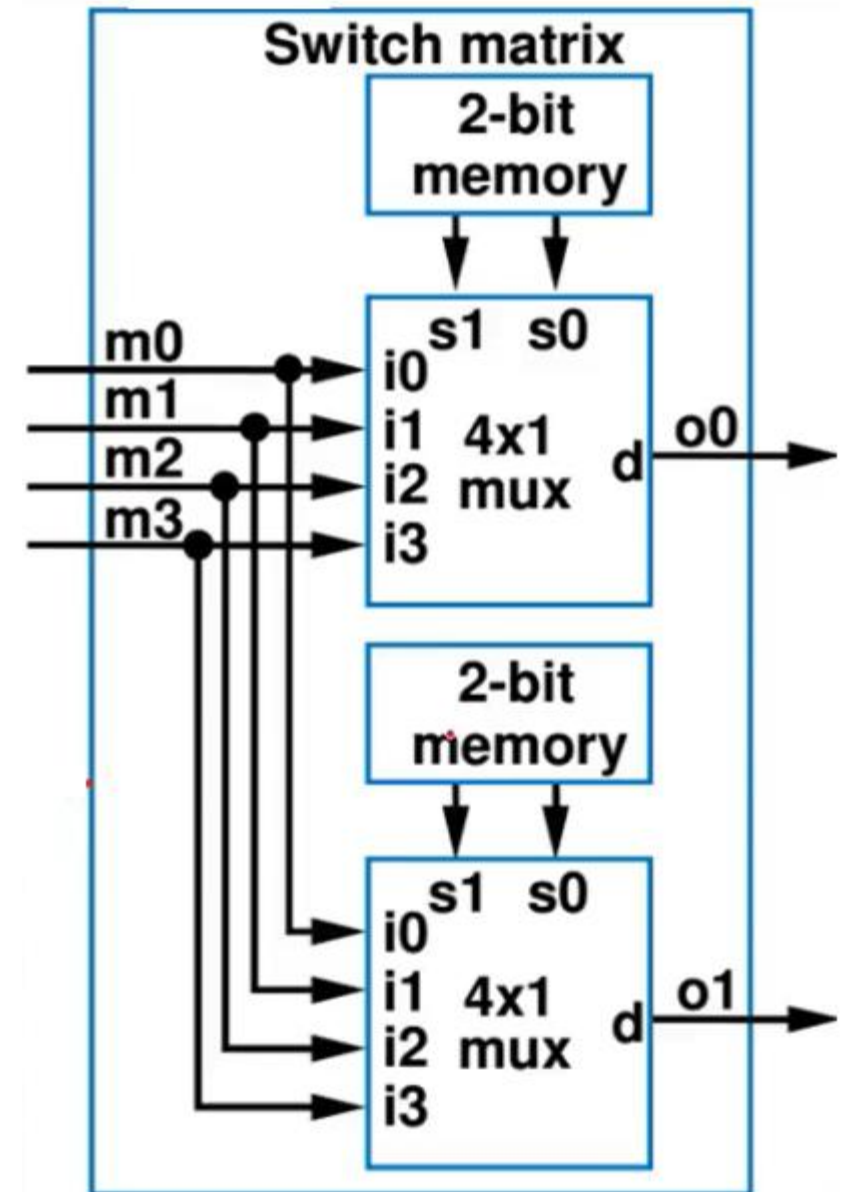
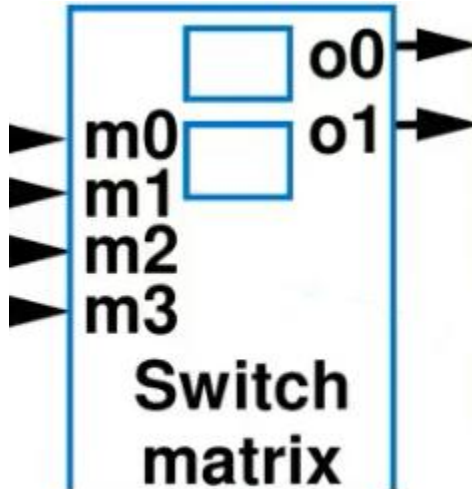
Truth Table					
A_1	A_0	D_3	D_2	D_1	D_0
0	0	0	0	0	1
0	1	0	0	1	0
1	0	0	1	0	0
1	1	1	0	0	0



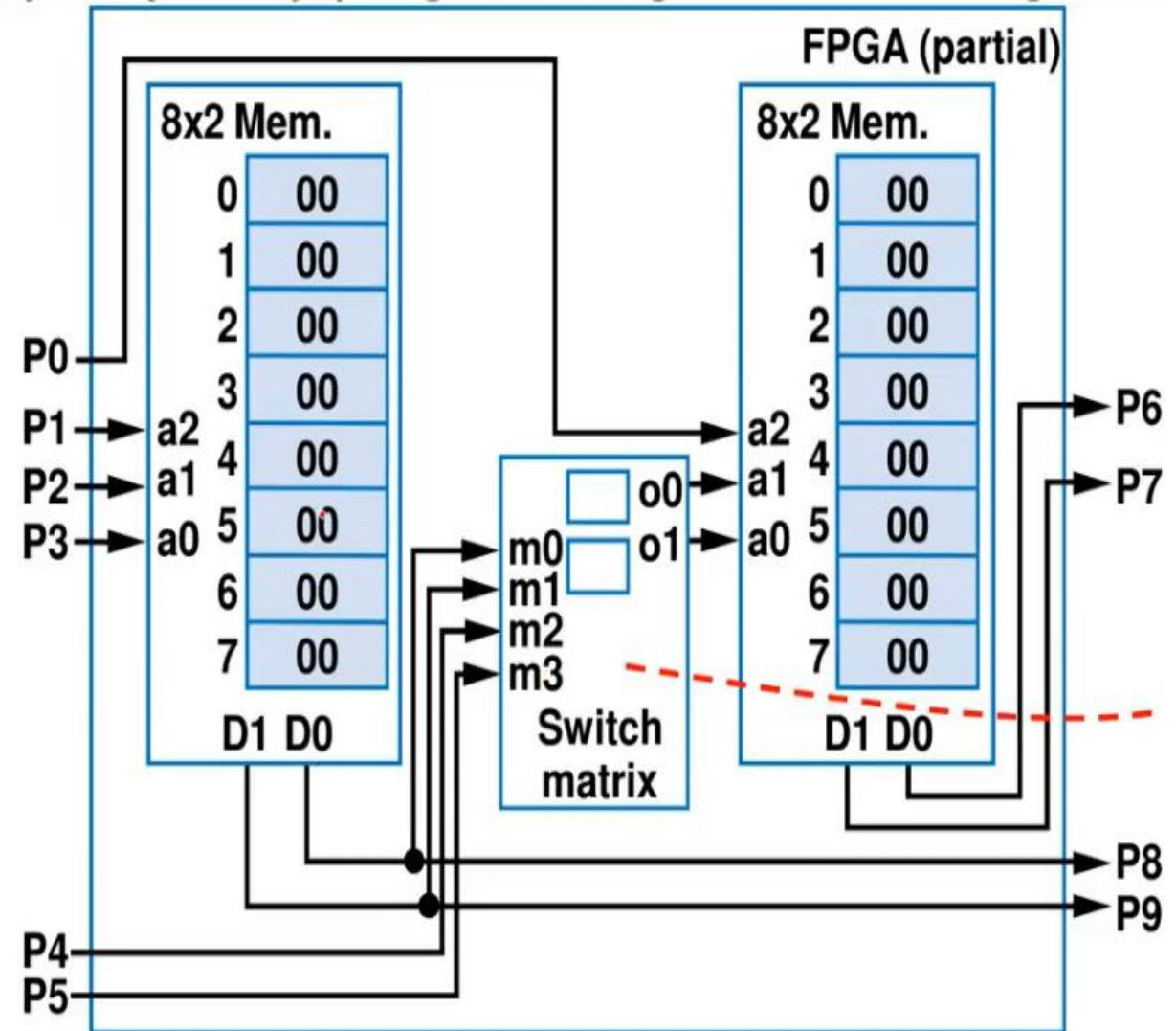
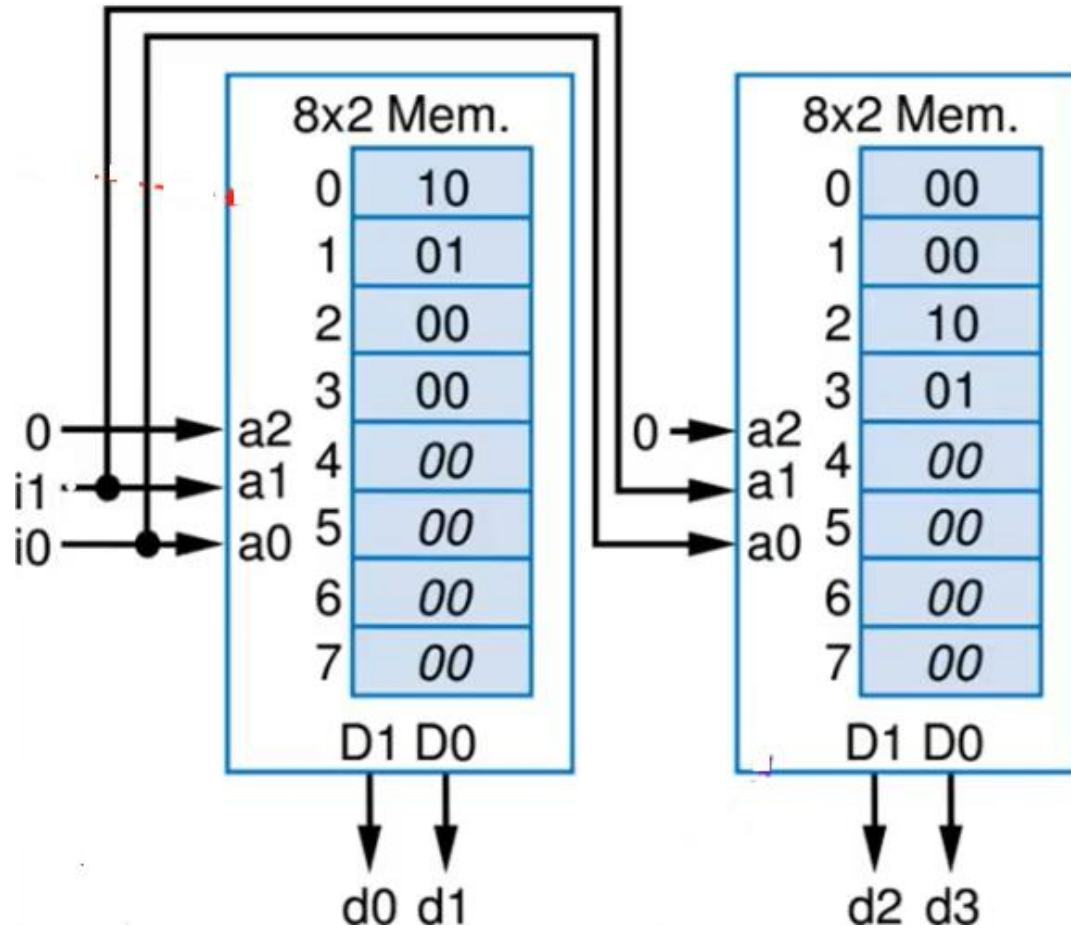
LUT



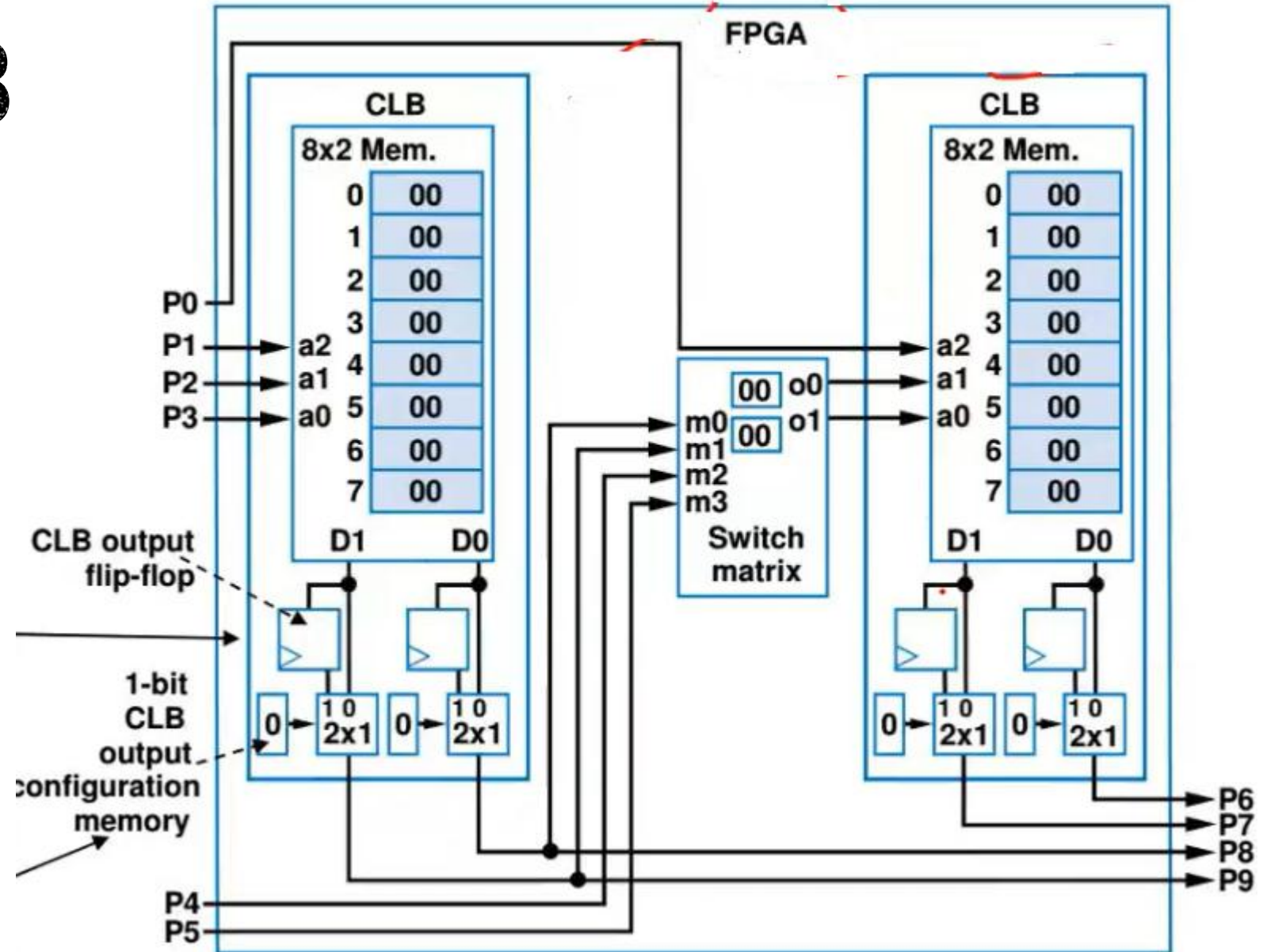
SWITCH MATRIX DESIGN??

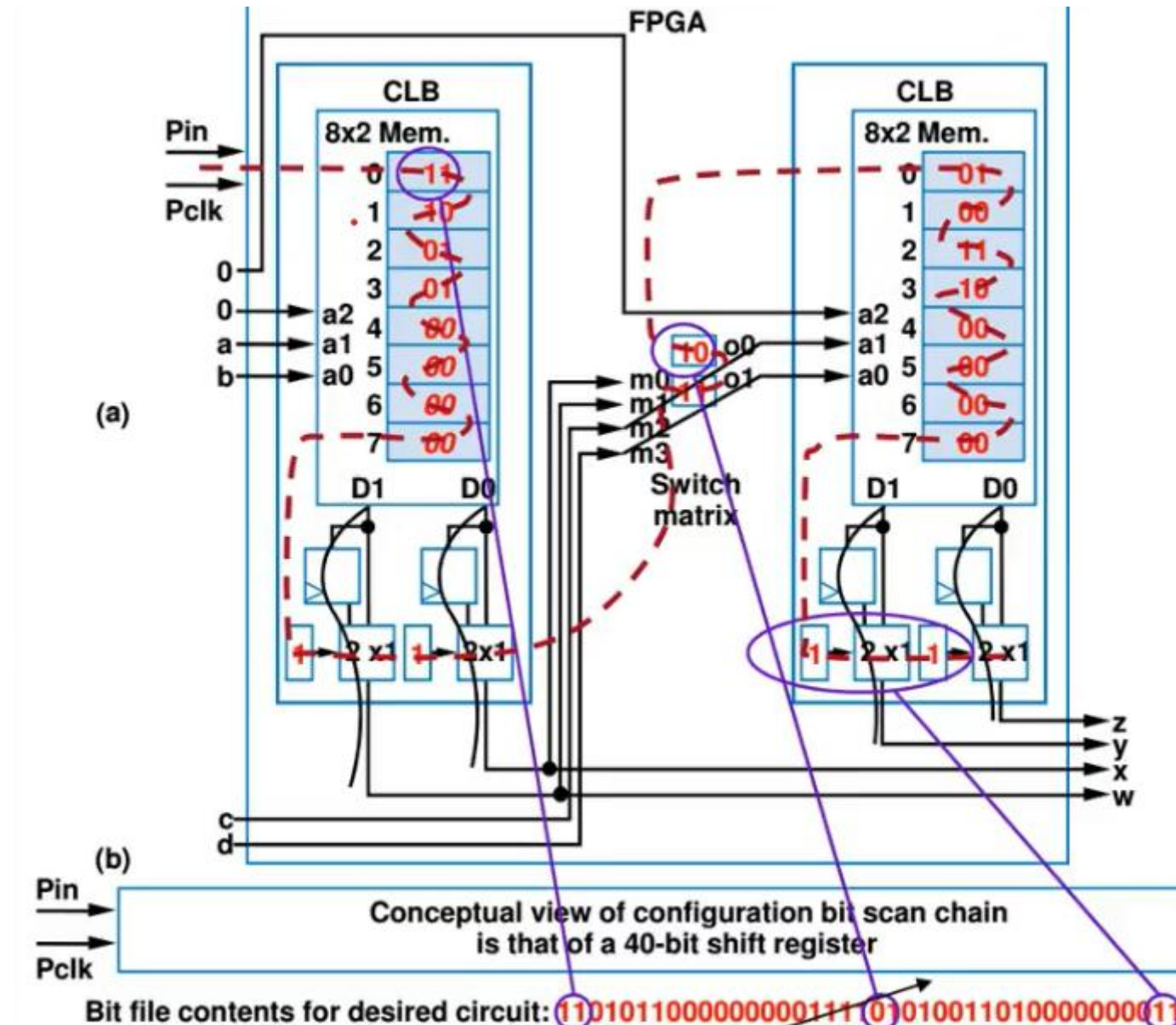


IMPLEMENT 2X4 DECODER USING 3-INPUT 2-OUTPUT LUTS

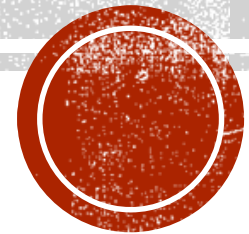


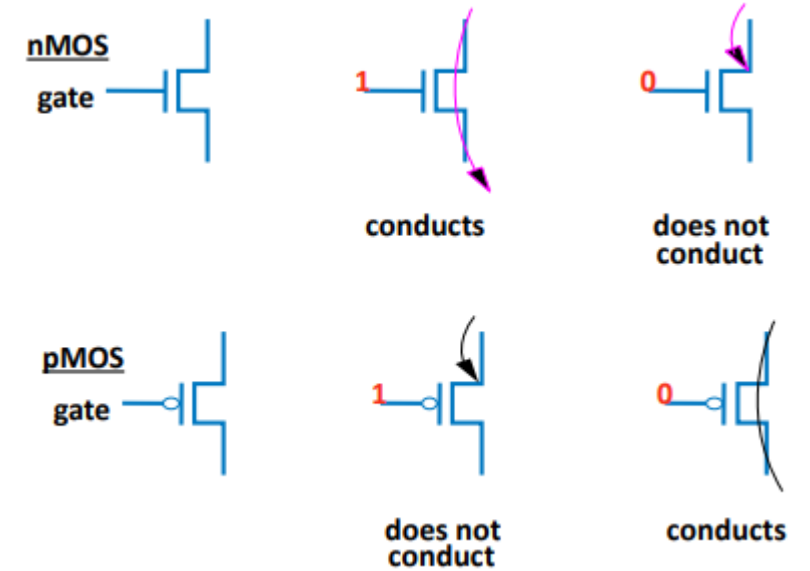
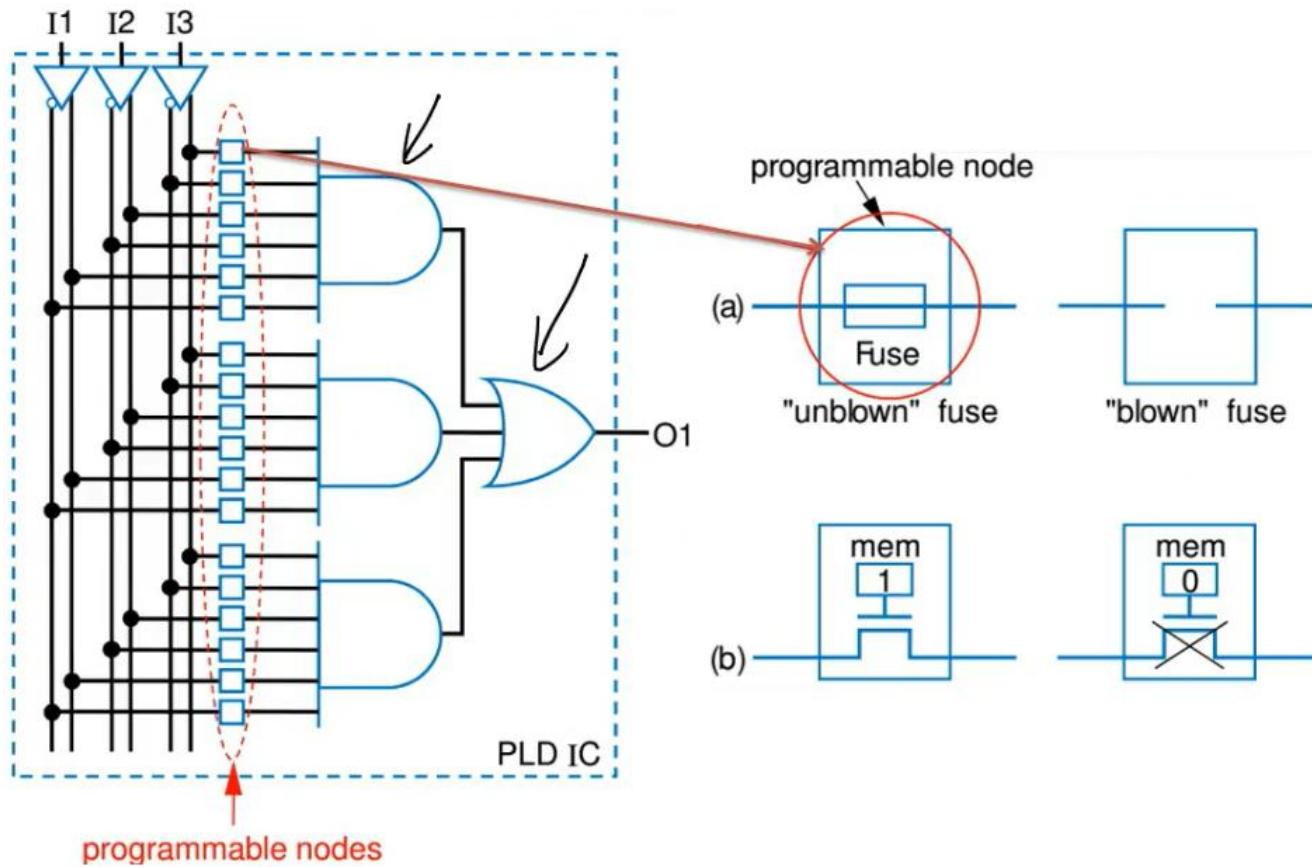
LUT > CLB



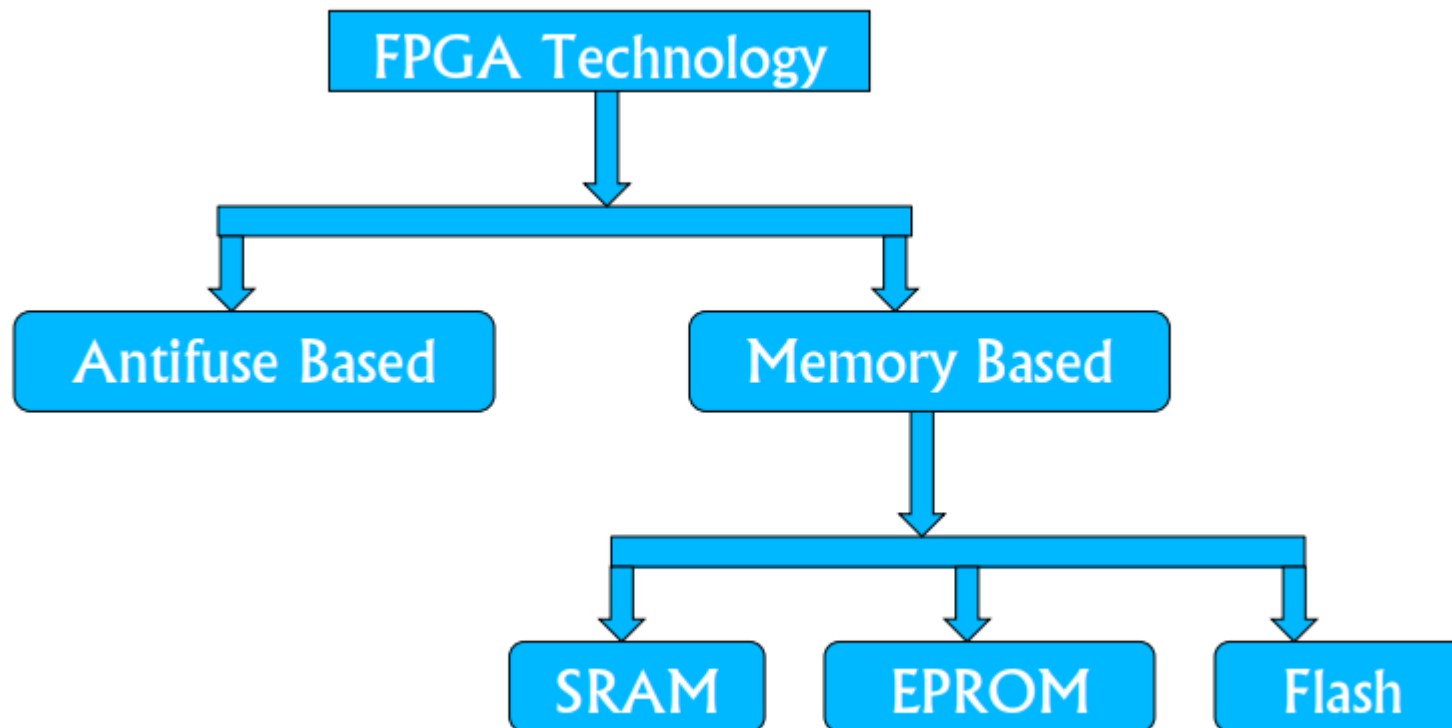


PROGRAMMABLE INTERCONNECT

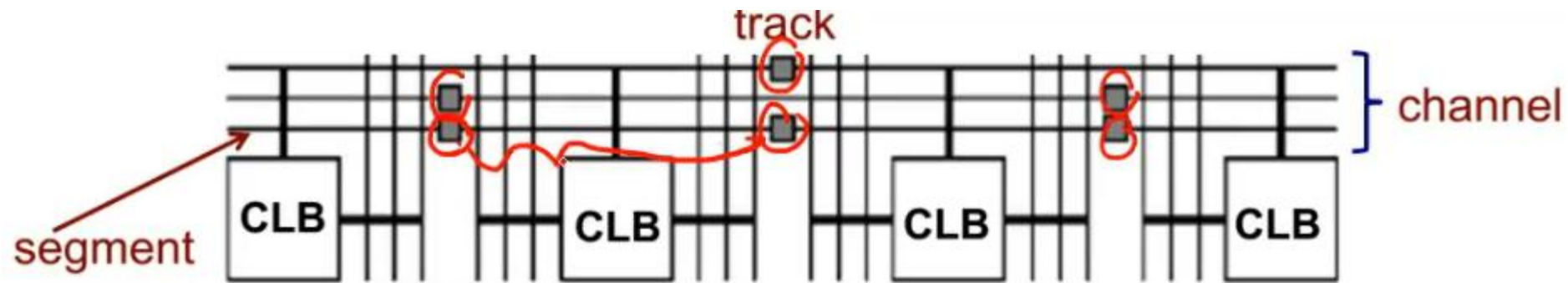




FPGA PROGRAMMING TECHNOLOGIES

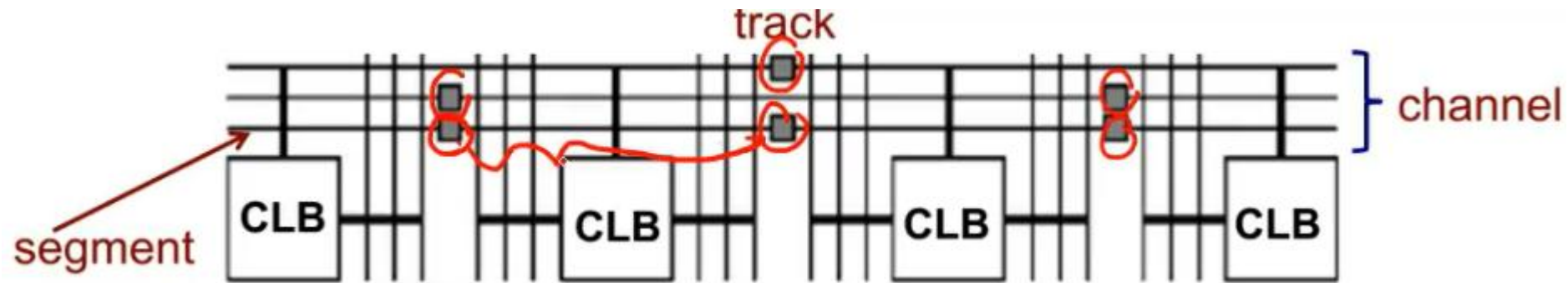


- Routing resources may take more area than logic



- A wire segment is a wire unbroken by programmable switches (can be of different lengths)

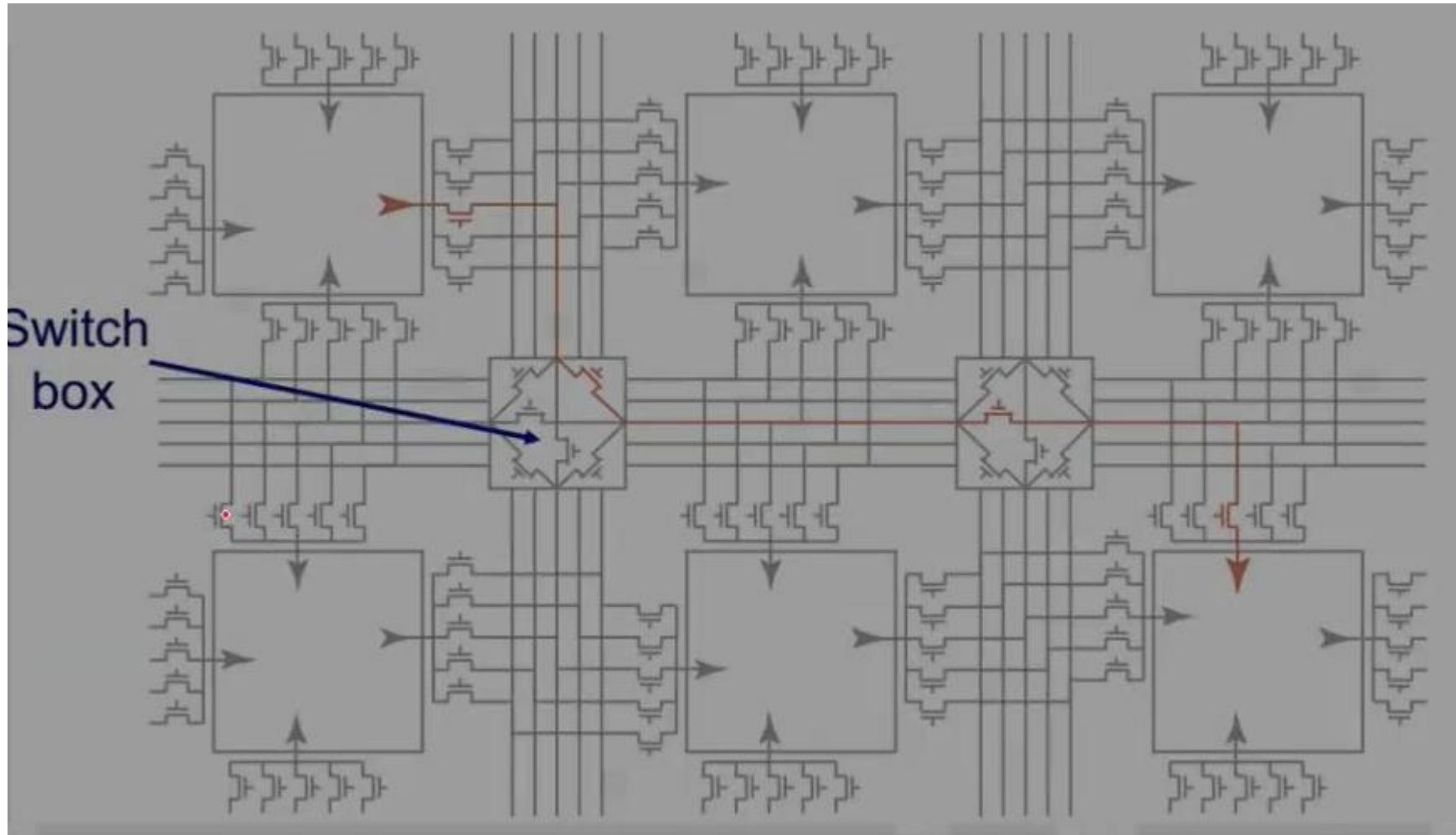




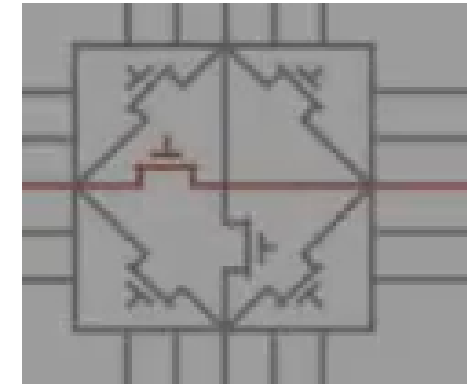
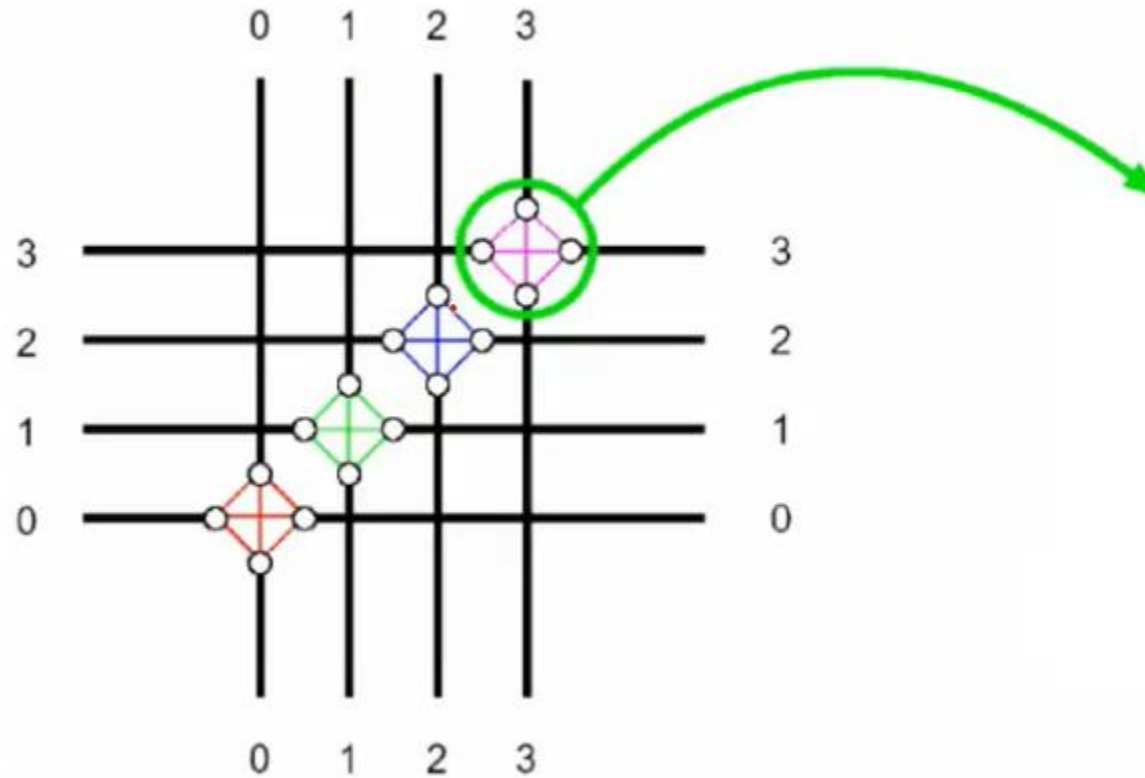
- A track is a sequence of one or more wire segments in a line. The segments could be connected by switches at their ends
- A routing channel is a group of parallel tracks. The channel width is the number of tracks in the channel.



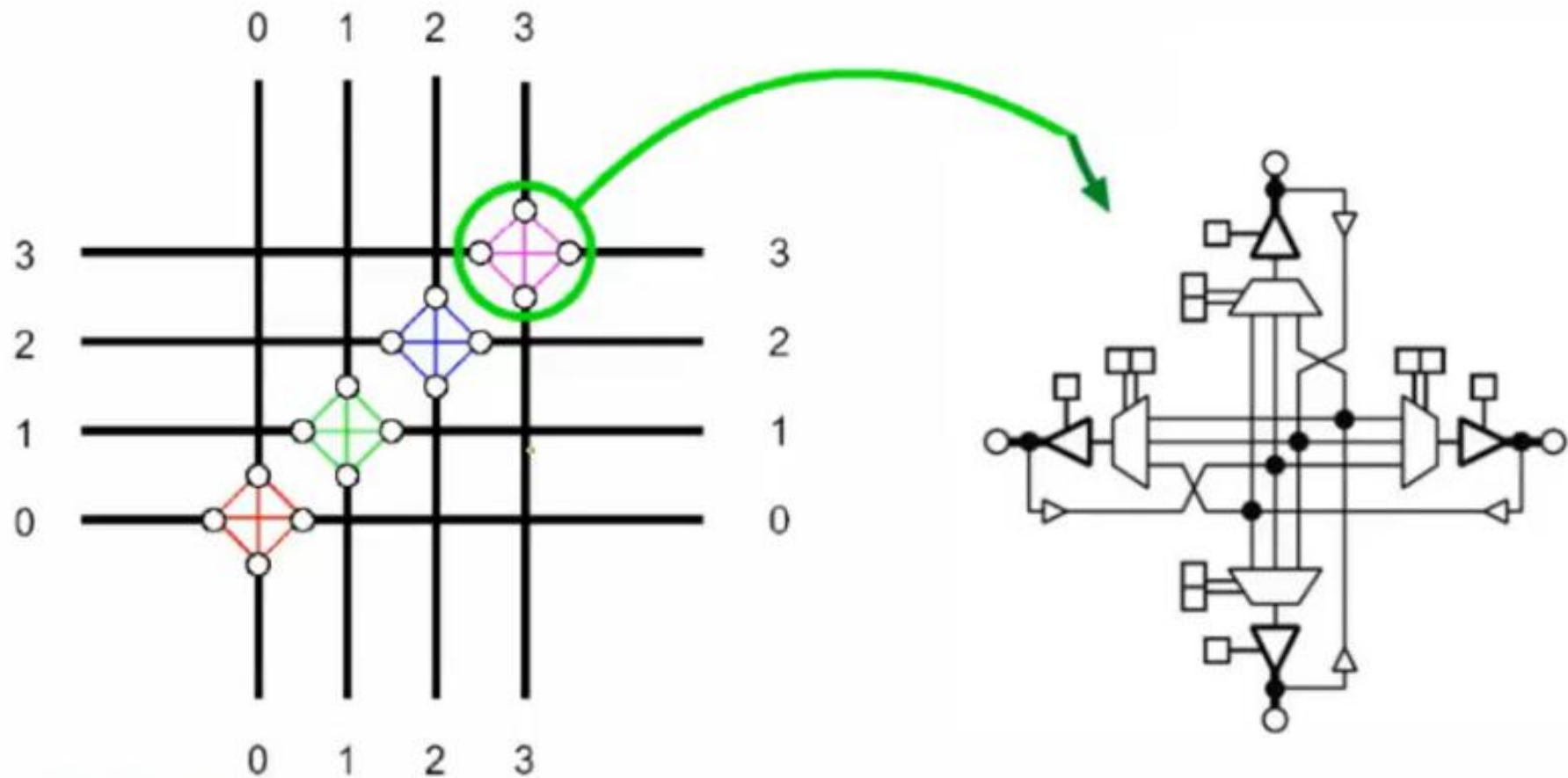
SWITCH BLOCK



BIDIRECTIONAL SWITCH DETAILS

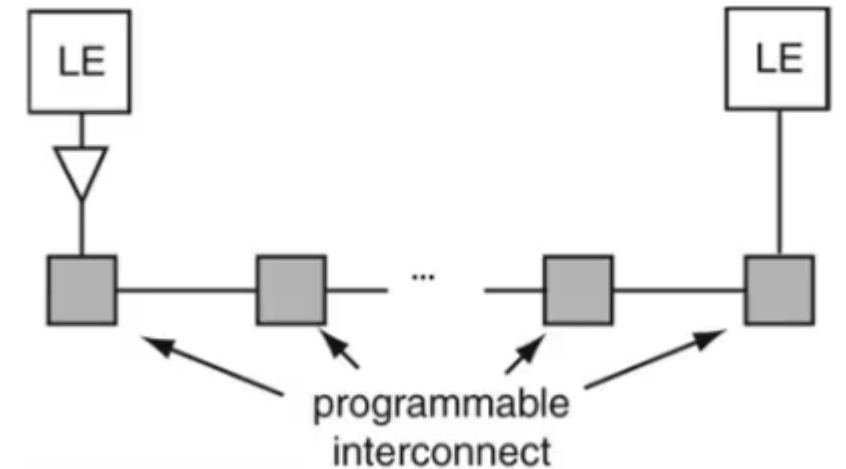


BIDIRECTIONAL SWITCH DETAILS



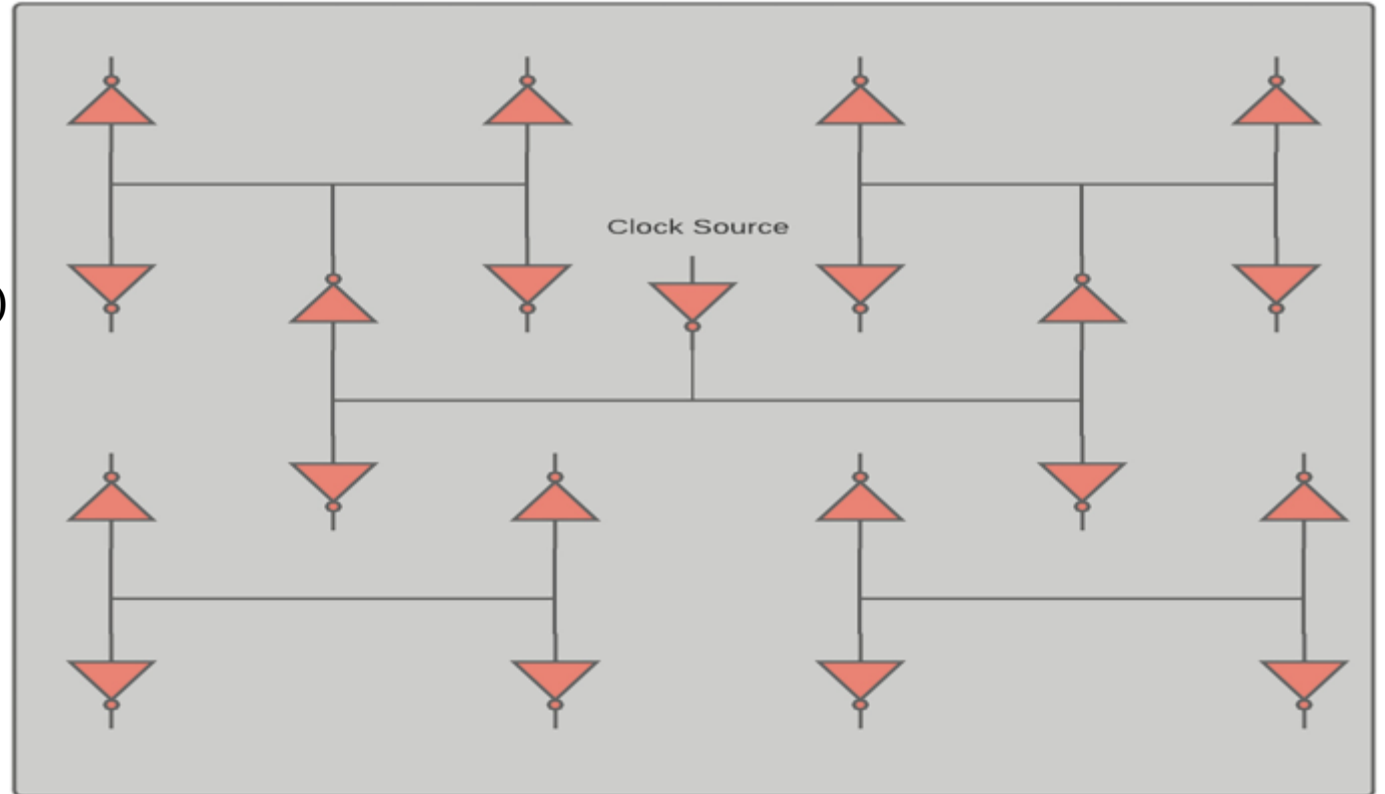
TYPES OF INTERCONNECT

- Short wires/Local Interconnect
- General Purpose interconnect
- Global Interconnects
- Specialized clock distribution

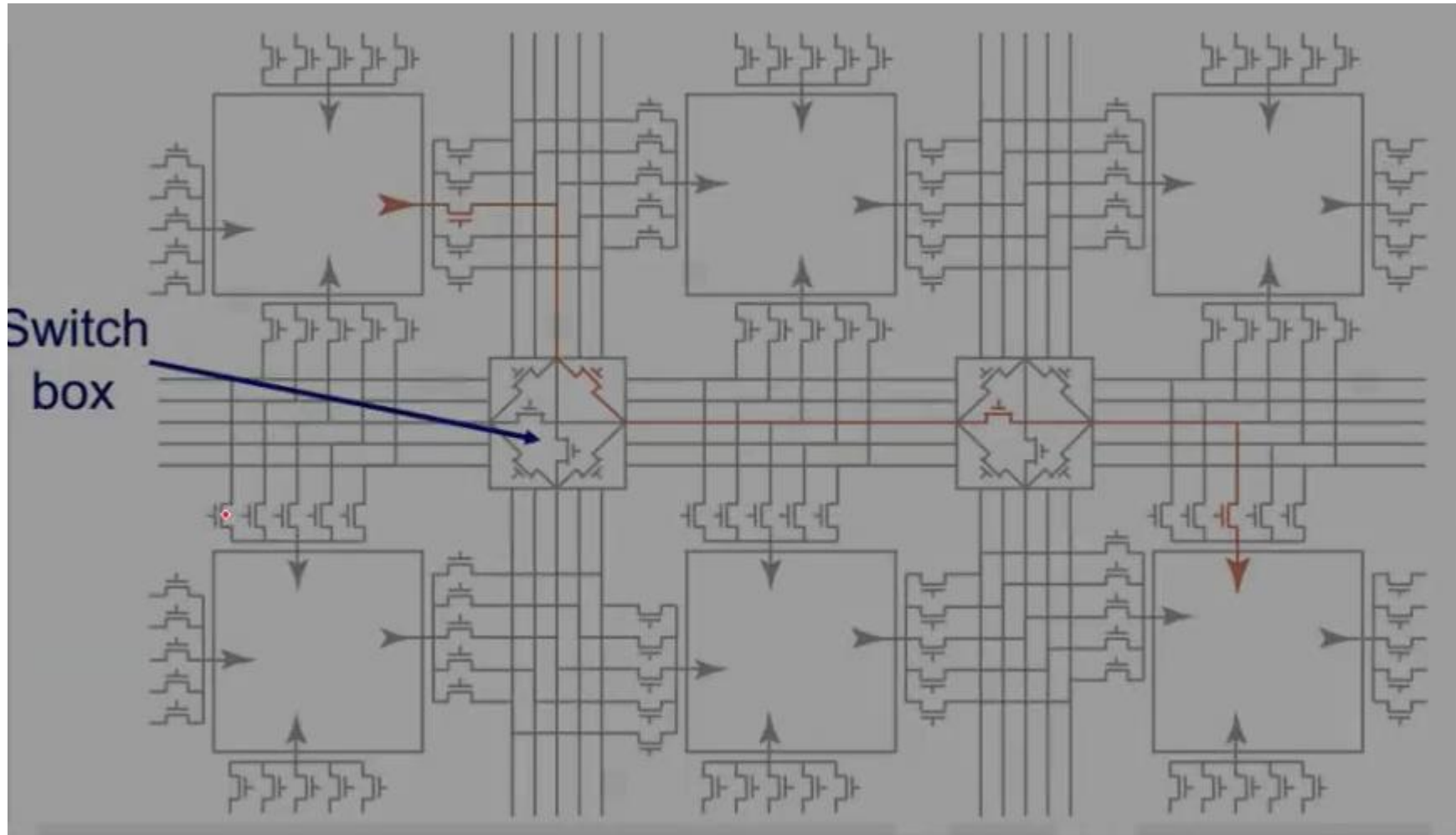


CLOCK DRIVERS

- Parameters
 - Number of fanouts
 - Load per fanout
 - Wiring tree parasitic (capacitance)
- Optimal Buffer sizes



SWITCH BLOCK



HIERARCHICAL

